

CPRE 431

Module 2 Lab

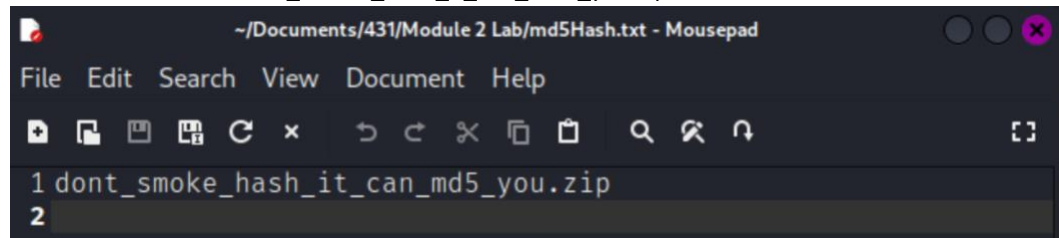
Assignments will be submitted in PDF format via Canvas.

Please submit your homework online through Canvas. Late homework will not be accepted.

Important: Your submission must be in .pdf format ONLY!

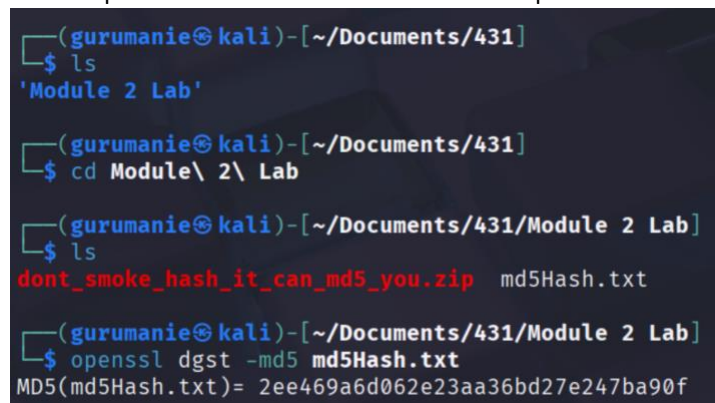
1. You are the security administrator in the XYZ company, and you noticed some strange traffic on the network of one of the employees sending an encrypted file to someone outside of the company. You want to check if the content doesn't have any confidential data. The encrypted file is attached with the homework (dont_smoke_hash_it_can_md5_you.zip). From the file name, you guessed that the password could be the md5 hash of the file name with the extension.
 - a. Explain with screenshots how to use OpenSSL tool to determine the md5 hash, then use any file decompression tool (WinRAR, Winzip, etc.) to decrypt contents of the attached file. Hint: write the file name in a text file then get the hash for this text file. Note that hash is space sensitive.

First, we create the txt file for the hashing process. The file is named md5Hash.txt has a single line with the contents as dont_smoke_hash_it_can_md5_you.zip.



Then we find the file in our directory and cd into where we have the txt file saved. Once we are there, we run the command mentioned: openssl dgst -md5 md5hash.txt

After the command runs, we get the hash of the file which is supposed to be the password. We use that password to extract the files from the zip file.



- b. After decryption, you noticed that the content of the attached file was an excel sheet “my_funny_expenses.csv”, an image “funny_image.png” and a text file “funny_jokes.txt”. This could look normal, however there is a lot of secrets and hidden messages in the content. Depending on your cryptanalysis skills, you started with the text file and you noticed that it only contained random characters but it started with the word “Salted__”. From your experience with OpenSSL tool, this means that this text file is encrypted using Symmetric Cipher (des, des3, aes128, aes192, aes256, etc.). Also, you noticed that all the three files have the word “funny” in their name, maybe this could be the encryption key. Explain with screenshots how to use OpenSSL tool to decrypt the original message in the text file.

```
(gurumanie@kali)-[~/Documents/431/Module 2 Lab/dont_smoke_hash_it_can_md5_you]
└─$ openssl enc -d -aes-256-cbc -in funny_jokes.txt -out un_encrypted.txt
```

Initially I had some trouble figuring out which exact openssl command to use to decrypt the funny_jokes.txt file. But after some trial I noticed that the above command works perfectly and is able to show the entire true contents of the file.

Given below are the true contents from the funny_jokes.txt file after decrypting using the above command.

```
1Hello there, as we agreed find attached the leaked new iPhone design schemes. Please transfer the money to my bank account. The bank account number and the instructions how to extract the design image are hidden in the excel sheet. Use "www.spammimic.com" to decode the excel sheet and get the info. Hint: copy the excel sheet using text editor (notepad++ or any similar), not with Excel processor.
```

- c. The decrypted message of part (a) will contain the instructions for this question. Explain with screenshots how did you solve this secret and get the new key.

After the previous step we go to www.spammimic.com and decode the excel sheet details. The below screenshot shows the result from decrypting the excel contents.



After putting the contents of the excel sheet into the decode spreadsheet function on www.spammimic.com we end up with the following message: Bank account: 112233445500. Use "https://www.mobilefish.com/services/steganography/steganography.php" to extract the design scheme from the attached image. Hint: use the bank account number as the password.

- d. The decrypted message of part (b) will contain the instructions for this question. Explain with screenshots how did you solve this secret and get the final leaked confidential information. Use the file on Canvas instead of the website given from the previous step to complete the final deliverable of the lab.

After part (c) gave us the Bank Account number and the website to extract the design from the png file, we go to that website and enter the password as the bank account number and upload the funny_image.png.

Decrypt: Unhide secret message or secret file from an encrypted image:

Upload encrypted image
Only *.png files (Max 4 MB)

funny_image.png

-- Or --

Enter image URL
Only *.png files (Max 4 MB)

Enter password:

To prevent automated submissions an Access Code has been implemented for this tool.

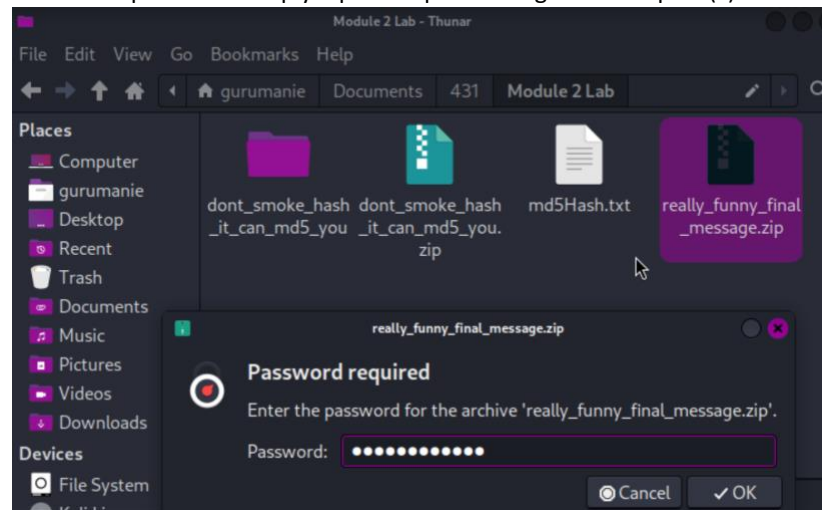
Please enter the Access Code as displayed above:

*** = required**

After using the above website to extract the design, we then download the extracted design through the website to get the following image:



Per the instructions, if we want to use the file attached in canvas instead, we can download that to our computer and simply input the password given from part (c):



This will give us a new extracted folder which contains the secret confidential image:

